

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1832

Pulaski 138 kV
October 04, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1832. The Interconnection Customer proposes a 76.8 MW battery energy storage facility interconnecting at the Pulaski 138 kV substation in the Pennsylvania area. The request was received on July 26, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 416 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,798,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Columbus Muni 1 / Pulaski 138 kV	Chase / Bryn Mawr 138 kV	124.9	201	100.0
Ascutney / Angora 138 kV	Bryn Mawr / Chase 138 kV	122.9	244	100.0
Bryn Mawr / Chase 138 kV	Ascutney / Angora 138 kV	119.2	251	100.0
Columbus Muni 1 / Ascutney 138 kV	Chase / Bryn Mawr 138 kV line	117.1	238	100.0
Guilbeau Electric / Oak Hall 138 kV L	Guilbeau Electric / Mount Holly 138 kV	113.1	362	8.1
Granite City / Bryn Mawr 138 kV	Ascutney / Angora Substation 138 kV Line	111.2	239	100.0
Lone Star / Moenkopi 138 kV	Plymouth Meeting / Barbadoes 138 kV	108.5	198	9.4
Lake Cowichan / Saville 138 kV line	26Th And F 138-1 kV Transformer	103.8	287	7.9
Granite City / Appleton 138 kV L	Ascutney / Angora 138 kV	102.4	223	100.0
Yreka / Tunnel 138 kV Line	Island Road / Subestacion Escuela li 138 kV	100.0	260	13.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Pulaski / Columbus Muni 1 138 kV line	Rebuild line with higher-capacity conductor	\$2,027,000	100% (\$2,027,000)
Ascutney Substation / Angora 138 kV	Rebuild line with higher-capacity conductor	\$6,707,000	100% (\$6,707,000)
Bryn Mawr / Chase 138 kV	Rebuild line with higher-capacity conductor	\$1,000,000	100% (\$1,000,000)
Ascutney Substation / Columbus Muni 1 138 kV Line	Rebuild line with higher-capacity conductor	\$1,129,000	100% (\$1,129,000)
Guilbeau Electric / Oak Hall 138 kV L	Rebuild line with higher-capacity conductor	\$3,190,000	38% (\$1,217,000)
Granite City / Bryn Mawr 138 kV	Rebuild line with higher-capacity conductor	\$2,560,000	100% (\$2,560,000)
Moenkopi / Lone Star 138 kV	Reconductor line	\$461,000	67% (\$311,000)
Lake Cowichan Substation / Saville 138 kV	Reconductor line	\$873,000	19% (\$170,000)
Granite City / Appleton 138 kV L	Reconductor line	\$276,000	100% (\$276,000)
Yreka / Tunnel 138 kV line	Reconductor line	\$887,000	45% (\$401,000)
	Total allocated to project		\$15,798,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.